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tl;dr

ML Engineering Manager and AI Hacker, solving the hardest problems by delivering bleeding-edge research into production. I have worked for 20+ years as a Researcher, Engineer, and Manager, tackling problems at every level: real-time scale, frontier innovation, and shipping usable products. My teams and I have consistently solved complex problems, owned a ton of infrastructure, and celebrated the delivery of ambitious projects.

ML + AI LLMs, Generative AI, AI Agents, RAG, MCP, Fine-tuning, PyTorch, vLLM, Evals, Knowledge Graphs

Engineering + Platform Terraform, Kubernetes, MLOps, Airflow, Kafka, Celery, AWS, Spark, Golang, Python

Achievements

- Released Clearwing and AGEST Zero Shield, an agentic SOC and pentesting suite
- Curated data collection and evaluation of unaligned models - covered in The Economist
- Fine-tuned Qwen3 for red-team use in 2025, before commodity cyber-specialized LLMs existed
- Built company-wide data pipeline and lake, unifying Research, Product, and Engineering
- Architected and deployed massive-scale, real-time data pipelines: 1.2M+ records/second
- Maintained infrastructure comprising 1,000+ instances

Education Bachelor of Science, Mathematics - Boston University

Work Experience

Lazarus AI — ML Engineering Manager — *October 2025 - Present*

Leading ML engineering and applied research across a broad portfolio: Cybersecurity, Public-Sector, Insurance. Here, my work has been delivering the latest AI technology into effective, comprehensible products for a wide range of contracts.

Stack: LangGraph, RAG, OCR, RLM, vLLM, Dagster, Langfuse, Axolotl, LLaMa-Factory, Python, Terraform, Kubernetes, MLOps

- Managed 3 concurrent project lines: Clearwing (Agentic SOC + Pentesting), AIE (Document Extraction), ATLS (Public Sector)
- Managed 4-person team, translating research into product tech stacks
- Built model-agnostic Agentic Systems across 4 projects, replacing legacy document extraction
- Extended document-parsing capability to 1,000+ pages at 1/10 per-page cost
- Migrated clients from cloud-hosted SaaS solutions to sovereignly-owned, on-prem solutions
- Synthesized and cataloged training data for the ReAligned/UnCut series of Lazarus Models
- Built LLM observability stack to run fully on-prem under HIPAA and SOC 2 constraints
- Drove evaluation-driven engineering culture, improving document extraction correctness from 20% to 85%
- Fine-tuned and deployed open-weight models on self-hosted inference infrastructure
- Built features for Clearwing, replicating 80% of project Glasswing's findings with open-weight models
- Owned data collection and evaluation for ReAligned, finetunes eliminating propaganda; covered in The Economist
- Owned production ML platform: k3s cluster, ArgoCD, Dagster pipelines

Warmer — Founding AI Engineer — *November 2024 - October 2025*

Building the future of Financial Advisory. In this role, I launched two AI products from zero to production: a financial advisor assistant and intelligent client-advisor matching system, optimizing Cost-per-Lead.

Stack: LangGraph, OpenAI, Python, LLM, LangSmith, PostHog, Celery, Amazon SQS, pgvector, RAG, GraphRAG, LightRAG

- Zero-to-One: founded and scaled AI product from zero to first paying customer
- Built conversational, lead-gen agent, the foundation of the new business model
- Achieved <1% hallucinations on golden-set evals through grounding and post-inference validation
- Architected multi-agent system managing complete client lifecycle, processing 1000+ meetings per day
- Introduced experiment-driven development culture with A/B testing and feature flags
- Built sub-12 second transcript analysis pipeline for real-time processing
- Designed complete agent architecture including RAG, memory, and tool usage systems
- Implemented hybrid-search RAG system across 1,000,000+ client communications
- Modified open-source dynamic Knowledge Graph, LightRAG, to support pgvector/Postgres backend

AI Consulting

AI Consultant and Contractor *September 2023 - November 2024*

Contracted with several early-stage startups, building out AI products.

Stack: LLM, TypeScript, LangSmith, Retrieval, Mixtral, HuggingFace, Python, LangChain, Twilio, Terraform

WorkMate Labs: Workmate is your AI teammate

- Designed and built a Digital Assistant, automating task extraction and email drafting
- Designed and deployed custom evaluation framework for measuring LLM performance
- Designed retrieval system for context-aware search of similar sent emails

Stealth Startup: FoodTech, translating conversations into API calls

- Fine-tuned Dolphin Mixtral to improve conversational style
- Designed and built question-answering workflows inspired by Graph-of-thoughts
- Automated MLOps for LLM-based projects - finetuning, testing, deployment, and performance evaluation

Stealth Startup: Generative AI Customers as a service

- Diagnosed and solved a mission-critical problem for scheduling flow of traffic to clients
- Designed and built API to develop, simulate, and actually orchestrate system traffic flows
- Standardized MLOps approach for use with Stable Diffusion and LLMs
- Designed and built tooling to monitor, compare, and research LLM performance

Owl.co

Software Engineer *January 2023 - September 2023*

Designed and built products for insurance tech, focusing on integrating ML into human workflows.

Stack: Clojure, PyTorch, Presto, Spark, Amazon SageMaker, Airflow, AWS CloudFormation

- Led an interdisciplinary team building ML products, automating tasks performed by human investigators: scraping the web, classifying documents, designing ETLs
- Automated hundreds of daily insurance claims investigations with the use of ML, reducing manual investigations by 70%
- Architected systems to integrate ML inference with human-driven tasks
- Designed and built ETL workflows for analytic databases
- Mentored mid-level Engineers on Data Engineering

Sabbatical

Travel, Meditation, Muay Thai *October 2021 – January 2023*

Oracle Data Cloud, MOAT

Software Engineering Manager *February 2017 - October 2021*

I led a highly technical team to create a vast, event-level data store, used as the source-of-truth for the suite of MOAT products. The real-time system processes 1.2M+ records/second, and requires zero downtime. Consequently, I grew a team with high technical aptitude, and emphasized ownership as a core principle in Software Development. Acquired by Oracle.

Stack: Kafka, Airflow, Parquet, Golang, Python, AWS, Protobuf, Postgres

Engineering Manager

- Managed and grew team of 7 Data Engineers, ranging from College Recruit to Senior Engineer
- Designed, built, and maintained infra end-to-end: web ingest, stream-processing, hourly ETL to Parquet
- Built multiyear Software Roadmap with Engineering Managers and Product Owners
- Designed and built the event-level data lake, unifying Research, Product, and Engineering datasets
- Managed a team owning 1,000+ instances with a budget of $340,000+ per month
- Collaborated with ML and Engineering to design a pipeline powering 5 distinct teams
- Mentored and promoted every Software Engineer on my team

Technical Lead

- Managed weekly software releases for core business logic, contributed to by 4 distinct teams
- Designed and built stream-processing applications processing 1.2M+ events/second
- Designed and built system-wide wire protocol
- Built custom software that reduced instance count by 50%, saving over $2M
- Maintained historical databases, importing 800,000,000+ rows per day - Highly modified Postgres

Chartbeat

Tech Lead, Data Engineering *December 2014 - December 2016*

- Led 8-person interdisciplinary team
- Designed core data pipeline, processing 300,000+ events/second - Kafka, Clojure
- Designed and built the first version of the Chartbeat Historical product, ranking millions of articles - Postgres

Prior Research

Harvard IQSS, NECSI *2005 – 2014*